

Software-centric Methodologies for T&M

A traditional test instrument is equipped with a fixed user interface and in-built programmed software and, hence, is not capable of testing multiple applications. This means that as soon as the test and measurement requirements change, the traditional test set-up has to be completely revamped. Let's take a look at another option



Modular instrumentation (Courtesy: www.ni.com)

ASHWIN GOPINATH

The most talked about feature for T&M systems nowadays is a software-based ecosystem. In the past, T&M was reliant on the traditional instrumentation approach for specific tasks. However, the testing capabilities of this approach have become increasingly inadequate due to the growing demand for instruments that support multiple tests. Software-defined, modular instrumentation is gaining popularity as it offers flexibility, scalability and higher speeds. A modular instrument is essentially generic hardware that acquires and

generates raw data while the software is stored on the user's PC, which allows user-defined measurements and analysis. A successful software defined T&M system comprises hardware platforms that are modular, reconfigurable and based on open industry standards, as well as system-design software that can be used to easily define the functionality of the system.

Chinmay Misra, technical marketing engineer, National Instruments India, explains the advantages of a software-centric ecosystem, "Since the instrument's functionality is software defined, clearly, a software centric ecosystem is of utmost importance.

Communities of developers and integrators, building on standard software platforms, using commercial off-the-shelf technology, make it possible to extend the functionality of complex hardware to cover applications previously impossible. With software easily available because of the ecosystem, it becomes possible for a test engineer to make a generic test system to test multiple products with the same hardware set-up. Using the same hardware keeps the costs down, while having an ecosystem makes IPs readily available and thus increases productivity." He explains that this is an example of application software made more valuable through its ecosystem.

A significant number of engineers have been trained on LabVIEW and have developed add-ons suitable for private application needs as well as other uses, through commercial vehicles like the system integrators in the NI Alliance Partner Network. In fact, an installable version of LabVIEW is being included with the September edition of EBY Plus magazine, so that interested engineers can test the software without having to download the huge file. With every additional supplier, producer, competitor or other stakeholder, the value of the software to each user, grows.

According to industry experts, the level of productivity and collaboration delivered by software-centric ecosystems will have a profound effect on test system design over the next five to ten years, and therefore all major T&M players are driving the growth of an